

MAT1100 Homework 2

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Question 3

Arithmetic for the permutations is done mod p unless explicitly stated otherwise.

- (a) *Proof.* Let p be prime and let $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ which is a field. Consider $\tau(x) = x + 1$ which is an element of $S_{\mathbb{F}_p}$ and let $P = \langle \tau \rangle$. Suppose we define $N = \{\sigma \in S_{\mathbb{F}_p} : \sigma P \sigma^{-1} = P\}$. To find all $\sigma \in N$ it suffices to find all $\sigma \in S_{\mathbb{F}_p}$ such that $\sigma \tau \sigma^{-1} = \tau^a$ as P is generated by τ . Note that τ has cycle representation $(0, 1, \dots, p-1) = (0, \tau(0), \tau^2(0), \dots, \tau^{p-1}(0))$ and so

$$\sigma \tau \sigma^{-1} = (\sigma(0), \sigma(1), \dots, \sigma(p-1))$$

now $\sigma(0)$ could be any element of $\mathbb{F}_p$. However it is clear that

$$\tau^a(x) = x + a$$

and hence τ^a has cycle decomposition $(0, ka, 2ka, \dots, (p-1)a) = (0, \tau^a(0), \tau^{2a}(0), \dots, \tau^{p-1}(0))$ so if

$$\sigma \tau \sigma^{-1} = \tau^a$$

we have

$$(\sigma(0), \sigma(1), \dots, \sigma(p-1)) = (0, \tau^a(0), \tau^{2a}(0), \dots, \tau^{p-1}(0))$$

and hence if $\sigma(0) = ka$ then $\sigma(1) = (k+1)a$. σ is now completely determined as this implies $\sigma(i) = (i+k)a$. Therefore all such possible σ are determined by whether they can map 0 and the possible values of a . $\sigma(0) \in \{0, \dots, p-1\}$ and $a \in \{1, \dots, p-1\}$ where $a \neq 0$ as no permutation is conjugate to the identity.

Now this implies that $|N| = p(p-1)$ with explicit representation

$$N = \{\sigma \in S_{\mathbb{F}_p} : \exists a \in \{1, \dots, p-1\} \text{ s.t. } \forall i \in \{0, \dots, p-1\}, \sigma(i+1) - \sigma(i) = a\}$$

i.e.

$$N = \{\sigma(x) = ax + b \in S_{\mathbb{F}_p} : a \in \mathbb{F}_p^\times, b \in \mathbb{F}_p\}$$

to show N is a subgroup of $S_{\mathbb{F}_p}$ it suffices to show that it is non empty and closed under the group operation as $S_{\mathbb{F}_p}$ is finite. Clearly $e \in N$ so $N \neq \emptyset$. Now if $\sigma, \rho \in N$ then

$$(\sigma\rho)P(\sigma\rho)^{-1} = \sigma(\rho P \rho^{-1})\sigma^{-1} = \sigma P \sigma^{-1} = P$$

and thus $\sigma\rho \in N$. Hence $N \leq G$.

Next we show that $P \triangleleft N$. Clearly $P \leq N$ as conjugating P by itself is an automorphism of P . Normality is clear as if $\sigma \in N$ then $\sigma P \sigma^{-1} = P$ by definition.

Now to characterize N further, note that the characterization

$$N = \{\sigma \in S_{\mathbb{F}_p} : \exists a \in \{1, \dots, p-1\} \text{ s.t. } \forall i \in \{0, \dots, p-1\}, \sigma(i+1) - \sigma(i) = a\}$$

gives a bijection of sets from N to $\mathbb{F}_p \times \mathbb{F}_p^\times$ given by the map

$$f : \sigma_{b,a} \rightarrow (b, a)$$

where $\sigma_{b,a}$ is the permutation given by $\sigma(x) = ax + b$.

Now I claim that $N \cong \mathbb{F}_p \rtimes_{\varphi} \mathbb{F}_p^\times$ with multiplication in the semidirect product defined as

$$(b, a) \cdot (b', a') = (b + ab', aa')$$

where $\varphi : \mathbb{F}_p^\times \rightarrow \text{Aut}(\mathbb{F}_p)$ is the identity as the automorphisms of $\mathbb{F}_p$ are precisely multiplication by a non-zero element (if ϕ is an automorphism it corresponds to multiplication by $\phi(1)$).

Now we need only show that the f above is a homomorphism as it is already clearly a bijection. First note that

$$\sigma_{b,a} \circ \sigma_{b',a'}(x) = \sigma_{b,a}(a'x + b') = (aa'x + ab' + b) = \sigma_{ab'+b,aa'}(x)$$

Therefore

$$f(\sigma_{b,a} \circ \sigma_{b',a'}) = f(\sigma_{ab'+b,aa'}) = (ab' + b, aa') = (a, b)(a', b') = f(\sigma_{a,b})f(\sigma_{a',b'})$$

as required. Now clearly we can identify P with the functions $x + b$ for $b \in \mathbb{F}_p$ therefore $P \cong \mathbb{F}_p$. Hence

$$N/P \cong \mathbb{F}_p^\times$$

□

- (b) *Proof.* $|S_{\mathbb{F}_p}| = p!$ and so $p \mid |S_{\mathbb{F}_p}|$ but $p^2 \nmid |S_{\mathbb{F}_p}|$ as otherwise this would imply $p \mid (p-1)!$ and since p is prime this means there exists $k < p$ such that $p \mid k$ which is clearly a contradiction.

Now $n_p = 1 \pmod p$ and furthermore P from (a) is clearly a Sylow p -subgroup of $S_{\mathbb{F}_p}$ with normalizer N . Therefore

$$n_p = [S_{\mathbb{F}_p} : P] = (p-2)!$$

hence $S_{\mathbb{F}_p}$ has $(p-2)!$ Sylow p -subgroups. Therefore

$$1 = (p-2)! \pmod p$$

and multiplying by $p-1$

$$p-1 = (p-1)! \pmod p$$

and hence

$$-1 = (p-1)! \pmod p$$

□

- (c) *Proof.* See (b).

□